Search Engine Marketing and Search Engine Optimization

MSIS 521

Professor Hossein Ghasemkhani

FOSTER SCHOOL OF BUSINESS
UNIVERSITY OF WASHINGTON

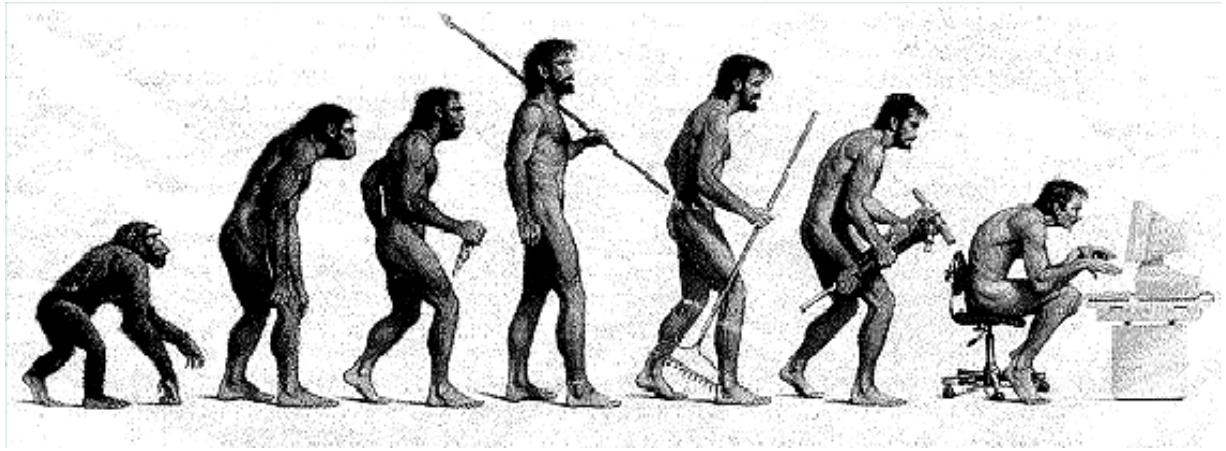
Learning Objectives

By the end of this session, you will be able to

- Explain how modern search results pages are structured
- Distinguish SEO vs SEM and when to use each
- Interpret key metrics (CTR, CPC, CPA, conversion rate)
- Understand how auctions & Quality Score affect ad position

FOSTER SCHOOL OF BUSINESS

Searching …



A large share of online sessions start with search

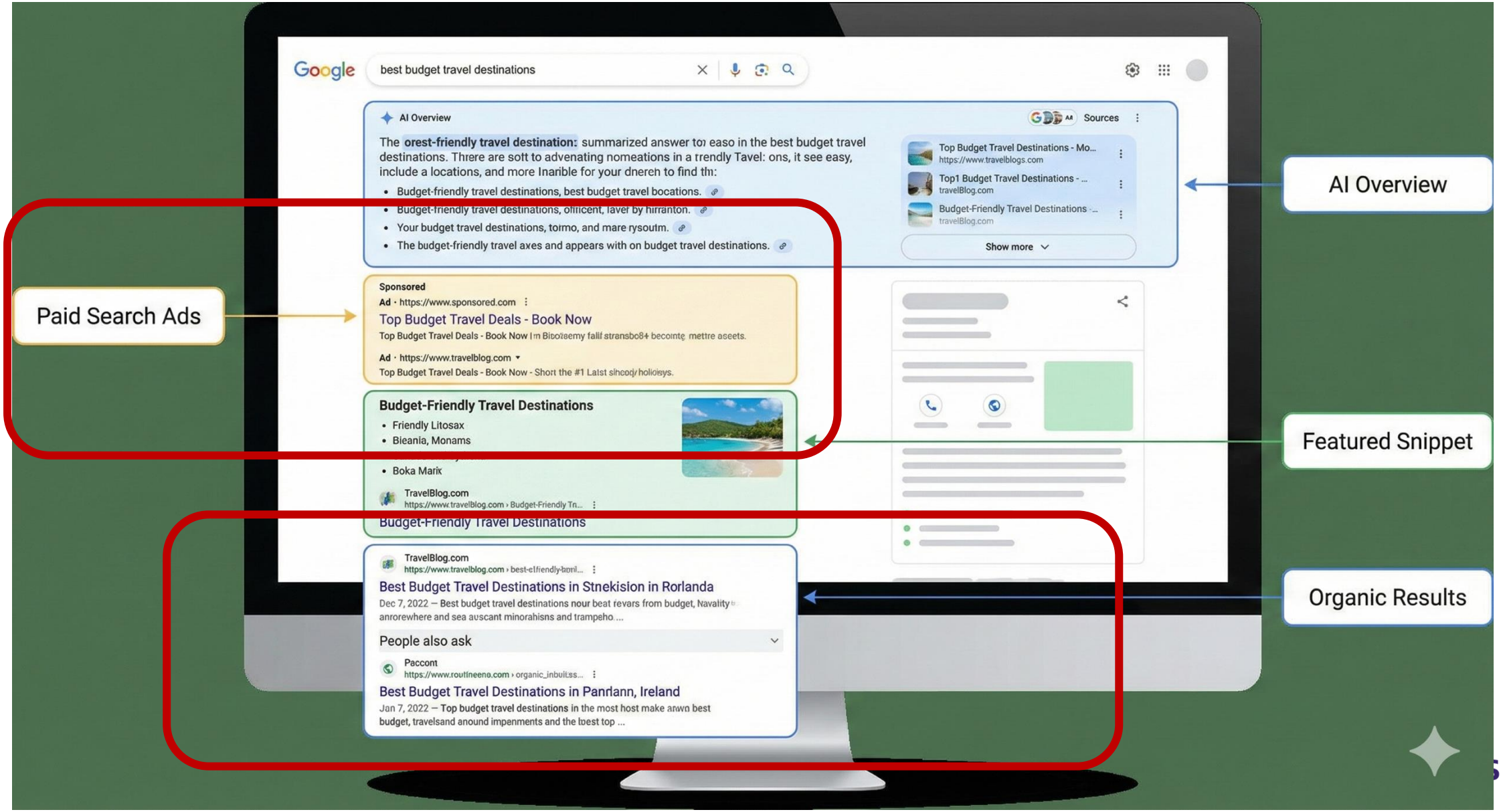
FOSTER SCHOOL OF BUSINESS

Why Search Matters?

- Think about search vs. other channels such as social, display, email
- Search is an intent signal

FOSTER SCHOOL OF BUSINESS

SERP (Search Engine Results Page)



How SERP has evolved

- Less blue links and more blended results
- Competition for attention above the fold (ads, snippets, etc.)
  - Between AI Overviews (AIO), Sponsored Ads, and Featured Snippets, the traditional #1 organic result is frequently pushed so far down that it's barely visible on mobile or laptop screens without a scroll.

FOSTER SCHOOL OF BUSINESS

How to Compete for Visibility in SERP

- The paid front: Google Ads
- The AI front: Optimizing for AI Overviews
  - The 60-word rule (structure the first 40–60 words of a section to be a definitive, extractable answer)
  - Information gain (publish unique data, case studies, or expert opinions that the AI can't just find in a dictionary)
- The structural front: featured snippets & PAA
  - Q&A formatting (using H2 and H3 headers as questions, following them with a concise answer)
  - Tables and lists (Google loves structured data!)
- The local pack:
  - Google business profile (focus on revies and local citations)

FOSTER SCHOOL OF BUSINESS

How Search Engines Work

Three main parts:

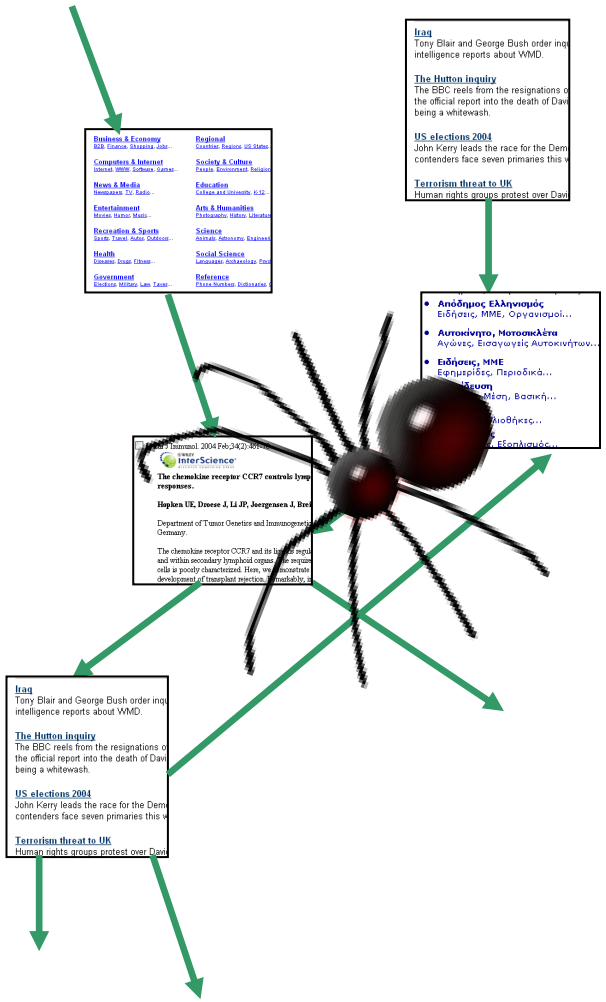
i. Gather the contents of all web pages (using a program called a **crawler** or **spider**)

ii. Organize the contents of the pages in a way that allows efficient retrieval (**indexing**)

iii. Take in a query, determine which pages match, and show the results (**ranking** and **display** of results)

FOSTER SCHOOL OF BUSINESS

How do Search Engines Discover Information?



- How do crawlers find web pages?
  - Start with a list of domain names, visit the home pages there.
  - Look at the hyperlinks on the home page and follow those links to more pages.
  - Keep a list of URLs visited, and those still to be visited.
  - Each time the program loads in a new HTML page, add the links in that page to the list to be crawled.

FOSTER SCHOOL OF BUSINESS

Indexing

Record the following information about each page

- List of words
  - Is the word in the title?
  - How far down in the page?
  - Was the word in boldface?
- URLs of pages pointing to this one
- **Anchor text** on pages pointing to this one
- And many other “secret ingredients” (**200+ factors** for Google)

FOSTER SCHOOL OF BUSINESS

How are Search Results Ranked?

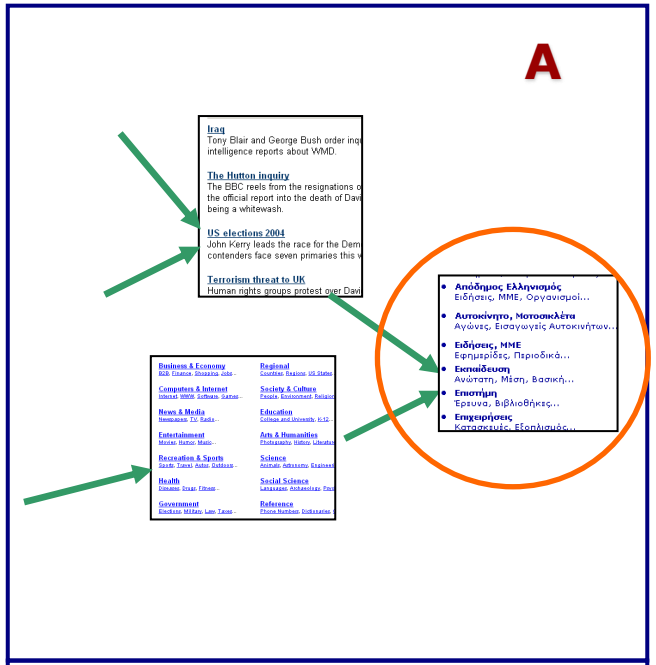
- Relevance
  - Content
  - Title
  - Headings
- Popularity and authority
  - links
- User behavior
  - Clicks
  - Dwell time

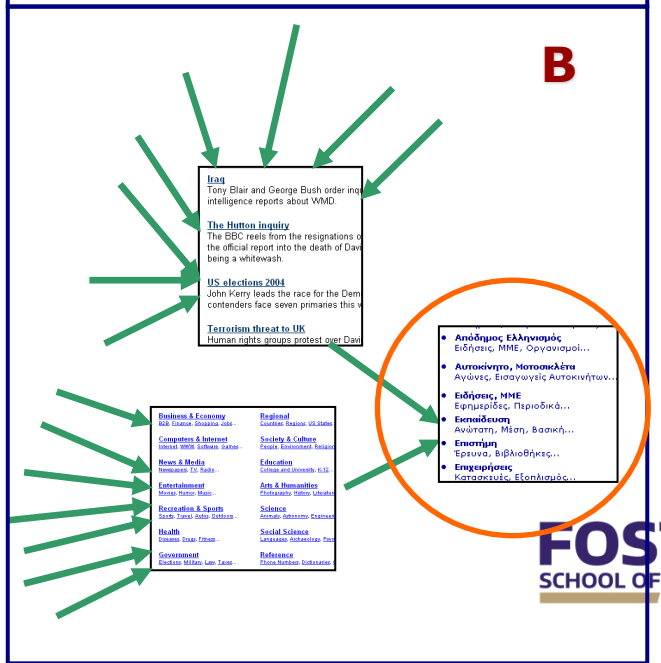
FOSTER SCHOOL OF BUSINESS

Measuring Importance of Linking

PageRank Algorithm

- Idea: important pages are pointed to by other important pages
- Method:
  - Each link from one page to another is counted as a “vote” for the destination page
    - The number of incoming links is important!
    - But it is not enough!
  - Each “vote” is different! Pagerank places more importance on votes that come from pages with a large number of votes (and so on, and so on)
- Compare, for example, the cases for the circled page in **cases A and B**

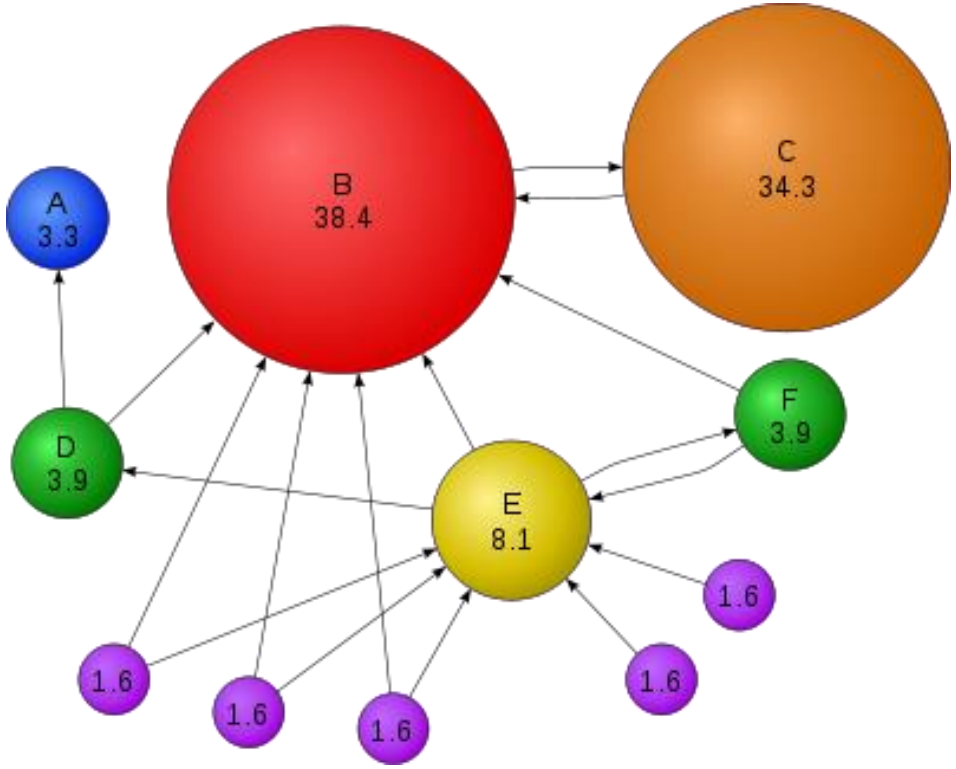




PageRank

- A variant of Eigenvector Centrality
- Damping factor d ~ 0.85
- Iterative computation

$$PR(p_i) = \frac{1-d}{N} + d \sum_{p_j \in M(p_i)} \frac{PR(p_j)}{L(p_j)}$$



Computing PageRank

(ignoring damping factor for illustration)

Page A

People who bought this also bought…

Page B

Page C

Page D

Page B

People who bought this also bought…

Page A

Page C

Page D

People who bought this also bought…

Page C

Page C

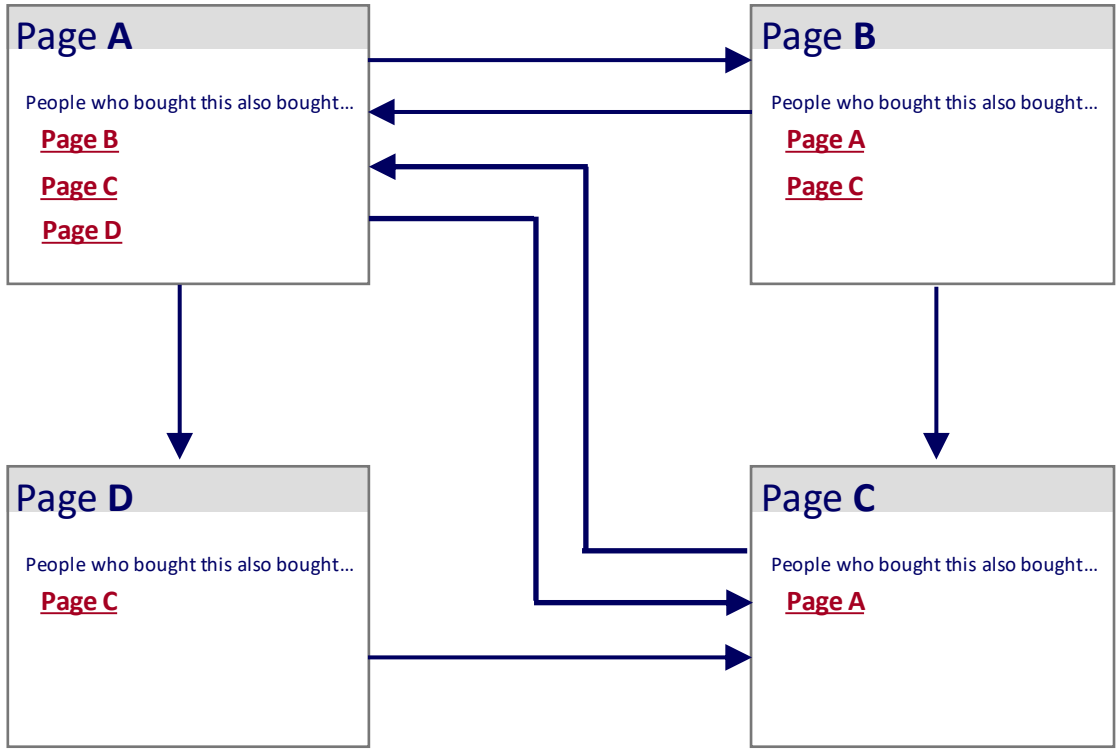
People who bought this also bought…

Page A

$$PR(i) = \sum_{j \in G(i)} \left(\frac{PR(j)}{OutDegree(j)} \right)$$

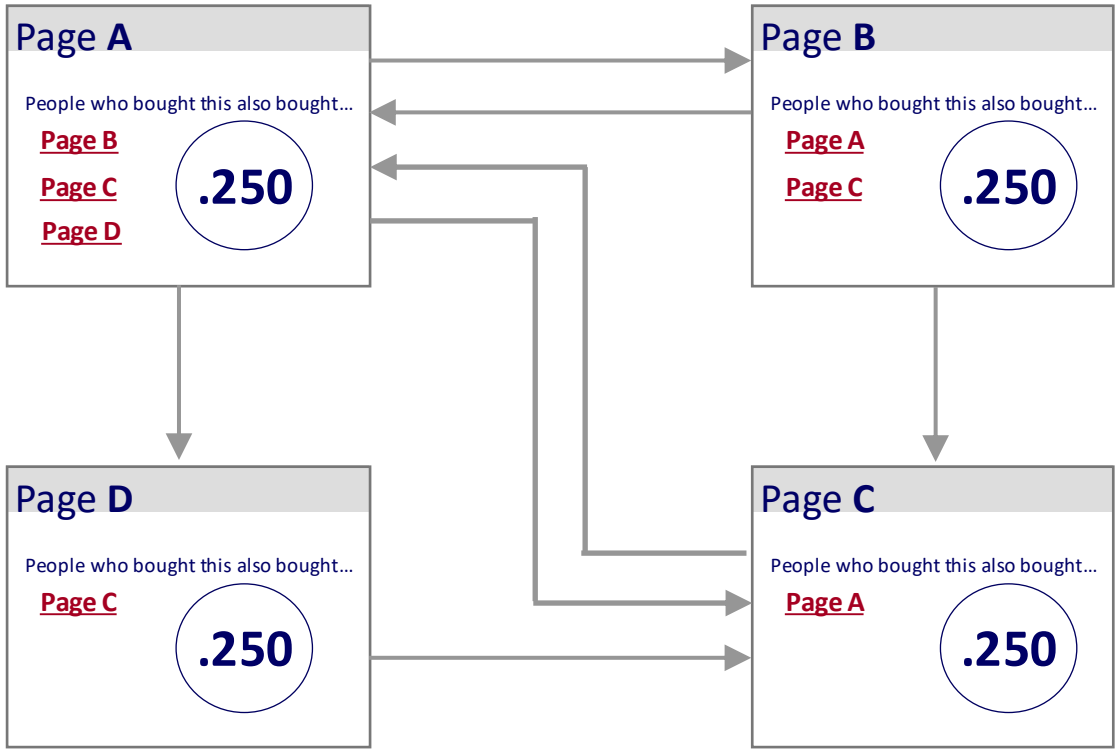
FOSTER SCHOOL OF BUSINESS

Computing PageRank



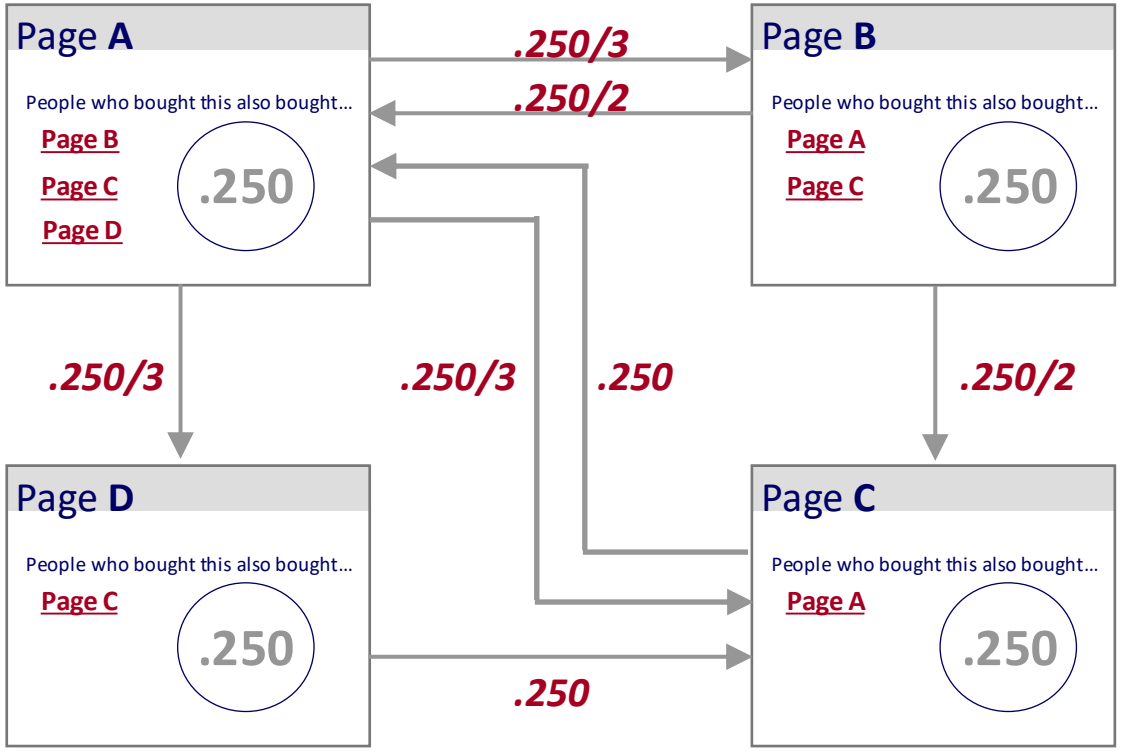
$$PR(i) = \sum_{j \in G(i)} \left(\frac{PR(j)}{OutDegree(j)} \right)$$

Computing PageRank



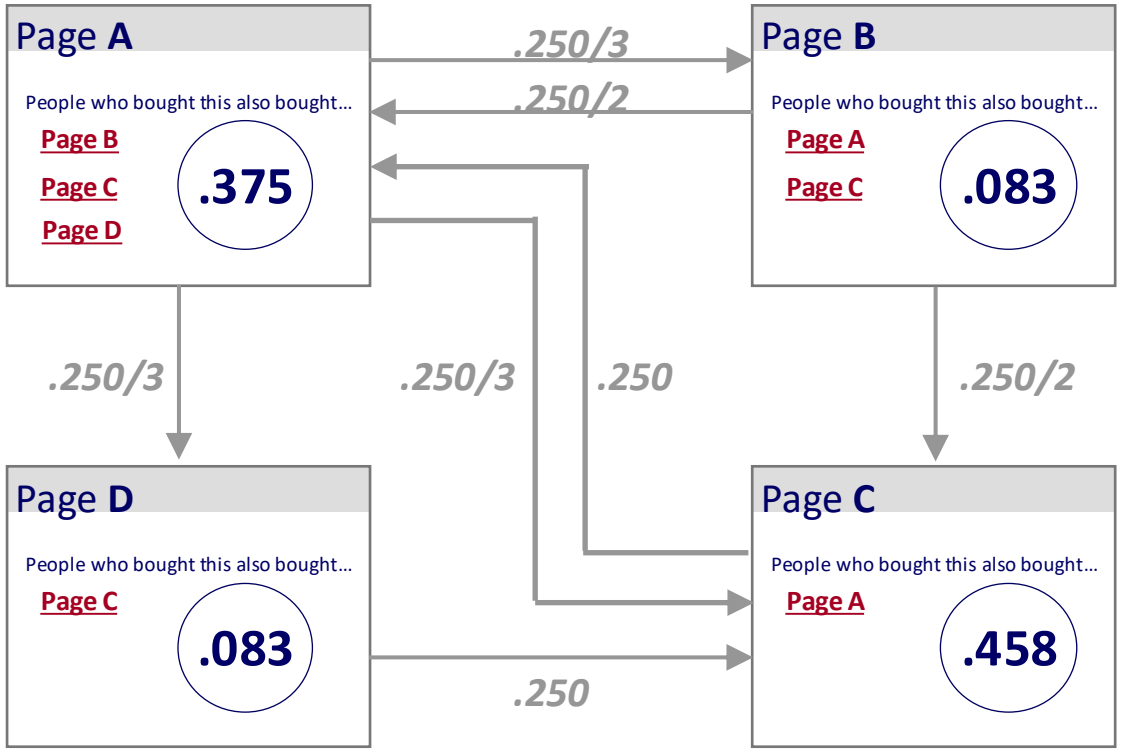
$$PR(i) = \sum_{j \in G(i)} \left(\frac{PR(j)}{OutDegree(j)} \right)$$

Computing PageRank



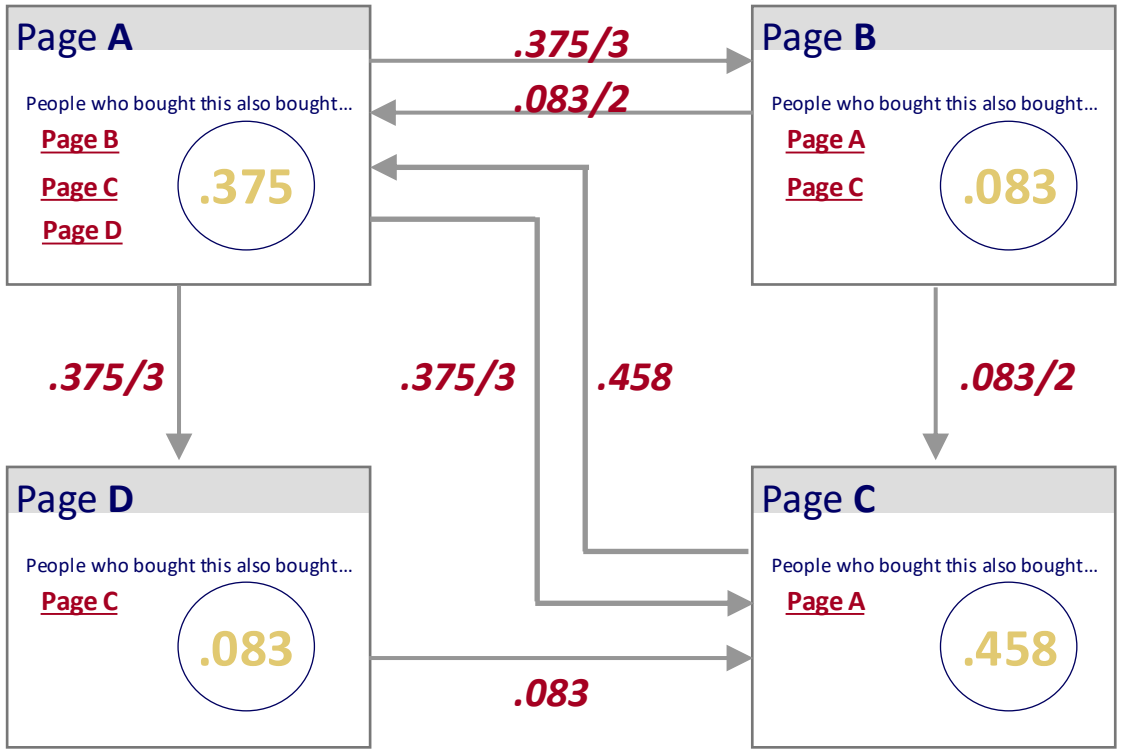
$$PR(i) = \sum_{j \in G(i)} \left(\frac{PR(j)}{OutDegree(j)} \right)$$

Computing PageRank



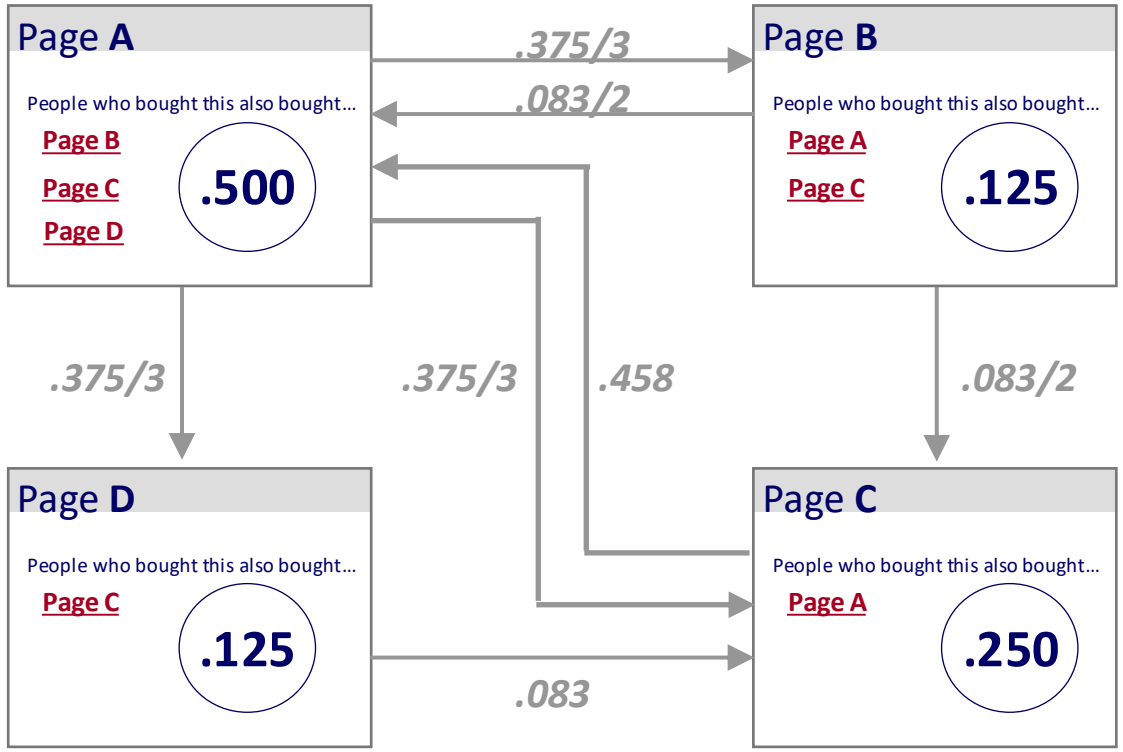
$$PR(i) = \sum_{j \in G(i)} \left(\frac{PR(j)}{OutDegree(j)} \right)$$

Computing PageRank



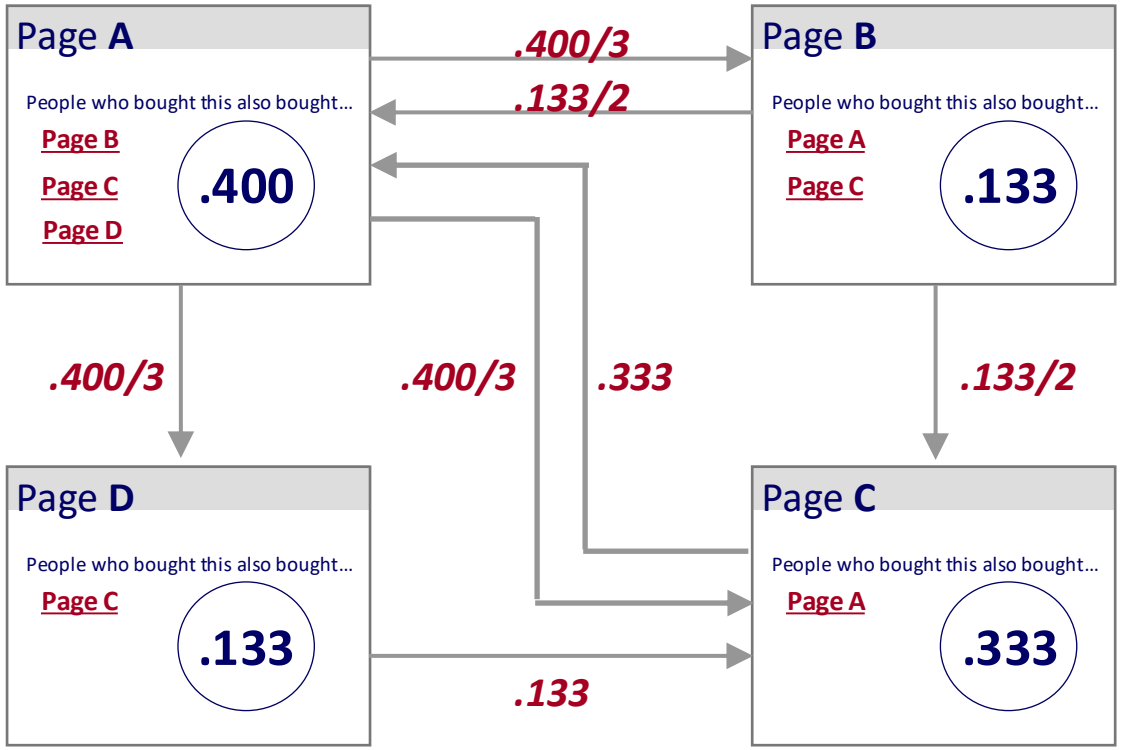
$$PR(i) = \sum_{j \in G(i)} \left(\frac{PR(j)}{OutDegree(j)} \right)$$

Computing PageRank



$$PR(i) = \sum_{j \in G(i)} \left(\frac{PR(j)}{OutDegree(j)} \right)$$

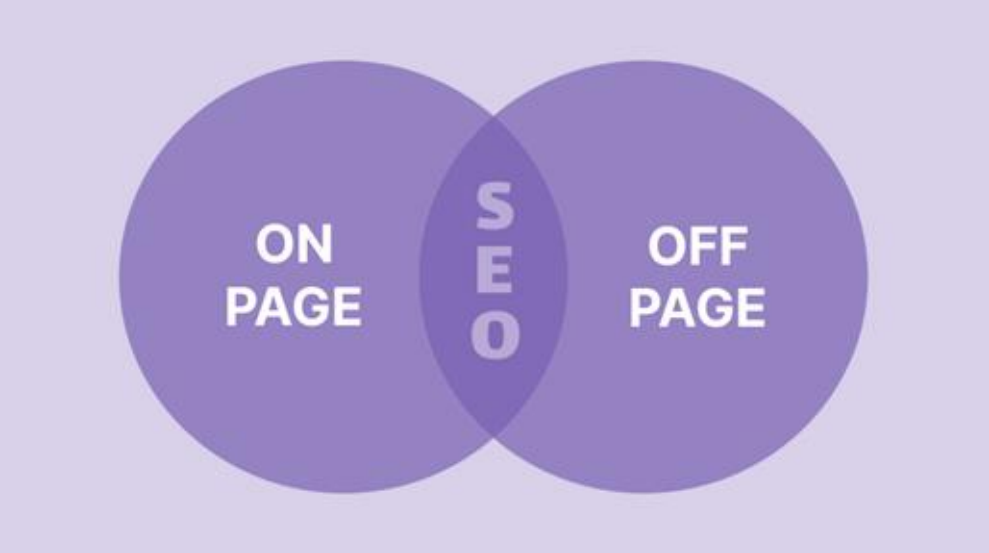
Computing PageRank



$$PR(i) = \sum_{j \in G(i)} \left(\frac{PR(j)}{OutDegree(j)} \right)$$

General Idea About How PageRank Is Used

- First the pages that contain the query text are located
- Weight the “text score” with the “link score”
- Rank results (using PageRank along with other factors like content and website quality and popularity)
- Search rank is very important as higher ranked results are much more likely to be visited!



FOSTER SCHOOL OF BUSINESS

What Marketers Can/Can’t Influence

- Levers marketers control:
  - Content quality and relevance
  - On-page structure (titles, headings, meta, internal links)
  - Link building & PR
  - Technical health (speed, mobile, security)
- What they can’t control:
  - Exact ranking formula
  - Algorithm updates & weights

FOSTER SCHOOL OF BUSINESS

What Is Search Engine Marketing (SEM)?

- **Search Engine Optimization (SEO)**
  - Improve "**organic**" result rankings on search engines
  - Adjust elements important to search engine algorithms
  - Avoid spamming techniques!

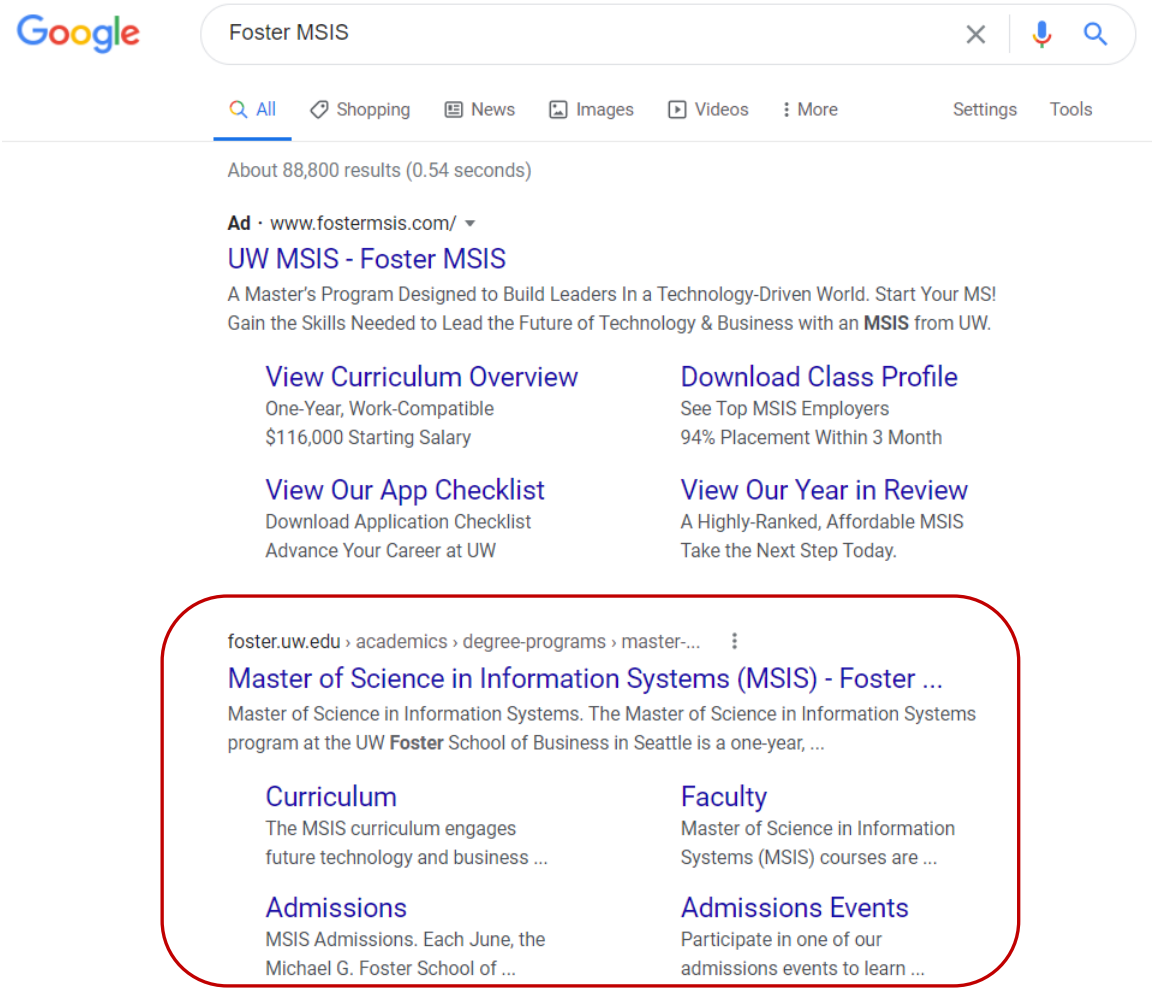
- **Pay Per Click Advertising (PPC)**
  - Also known as "**sponsored** results", advertising that appears to the top, bottom, or side of search engine natural results
  - Typically, cost is generated on a "pay-per-click" model, where advertiser pays for each click a searcher makes on an ad, vs. the traditional online marketing method of "cost per impression".

FOSTER SCHOOL OF BUSINESS

What Is SEO?

- SEO = Search Engine Optimization
- SEO is the process of optimizing/increasing your rankings in the Search Engine Results Page (SERP).
- Push vs Pull Marketing

https://www.salesforce.com/marketing/push-vs-pull

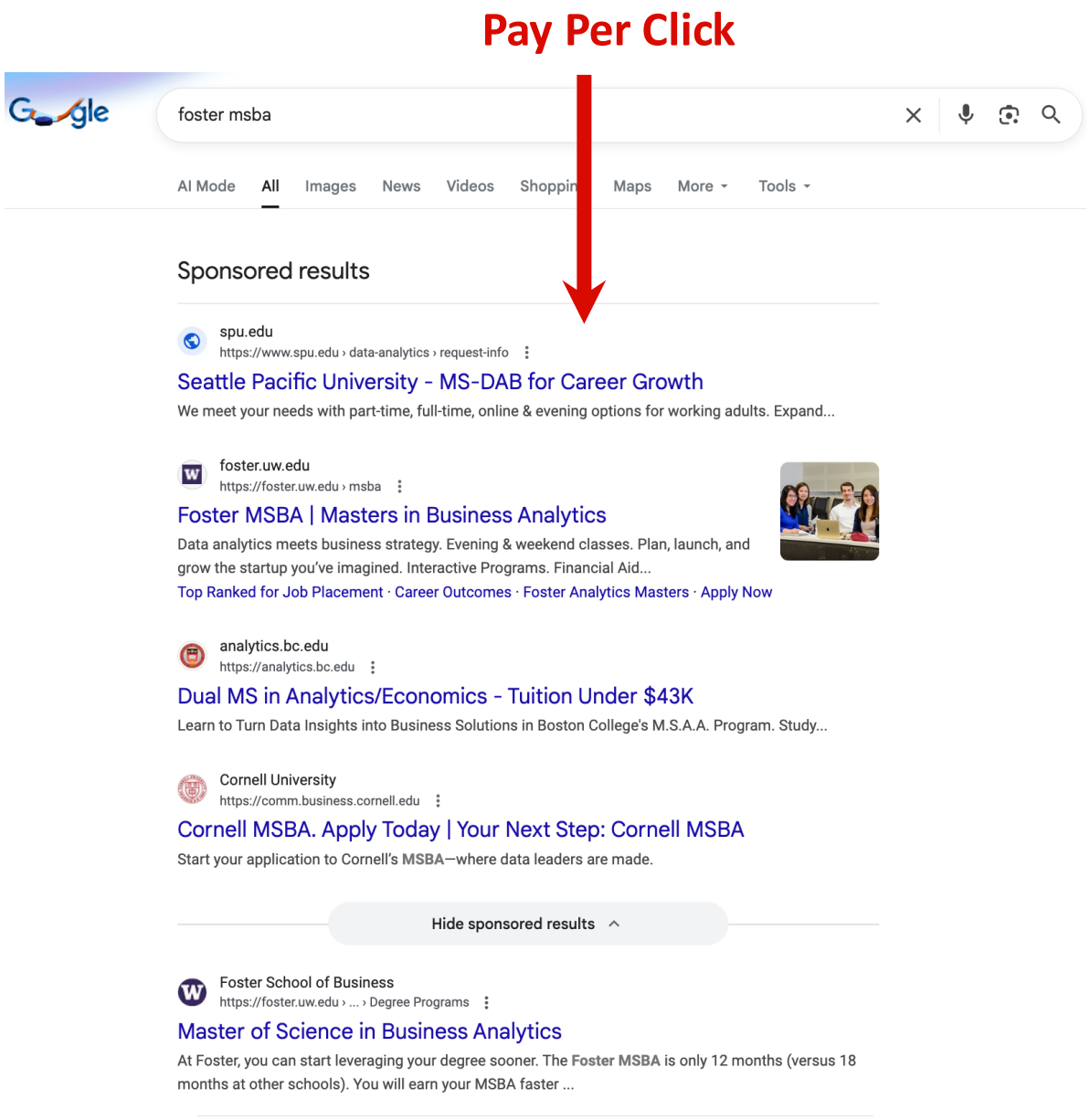


Search Engine Optimization

FOSTER SCHOOL OF BUSINESS

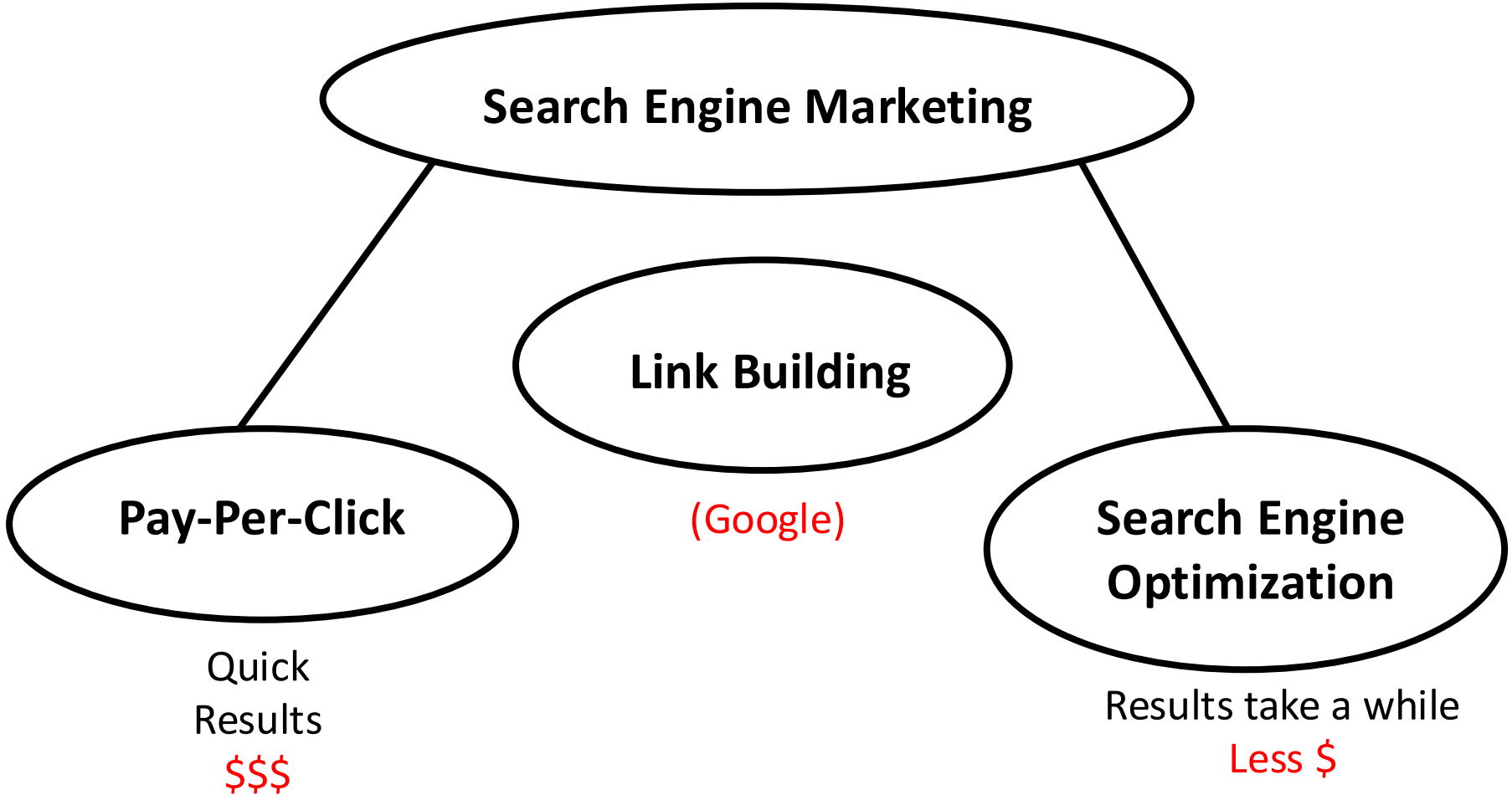
What Is Pay Per Click?

- Paid Advertisements on the top, bottom or right “Ads”, “Sponsored” or “Suggested Links”
- Con: fewer users click on paid ads
- Con: No sustaining or residual value (ads stop showing as soon as you stop paying.)
- PRO: Get results FAST!



FOSTER SCHOOL OF BUSINESS

How Does It Fit Together?



FOSTER SCHOOL OF BUSINESS

Some Best Practices for Traditional SEO

- On-page SEO (What you offer)
  - Titles, H1/H2, keywords use, informative content
  - Image ALT text, structured data basics
- Off-page SEO & authority (Who vouches for you)
  - Backlinks from other sites, anchor text, topical relevance

FOSTER SCHOOL OF BUSINESS

Technical SEO in 2025

How your website works behind the scenes.

- Focus on how the site is built and delivered, not just what it says
- Make pages crawlable & indexable (robots.txt, sitemaps, clean URLs)
- Ensure a fast, mobile-first, secure experience (Core Web Vitals, HTTPS)
- Use structured data to help search engines understand your content
- Monitor issues in Google Search Console (coverage, page experience, errors)

FOSTER SCHOOL OF BUSINESS

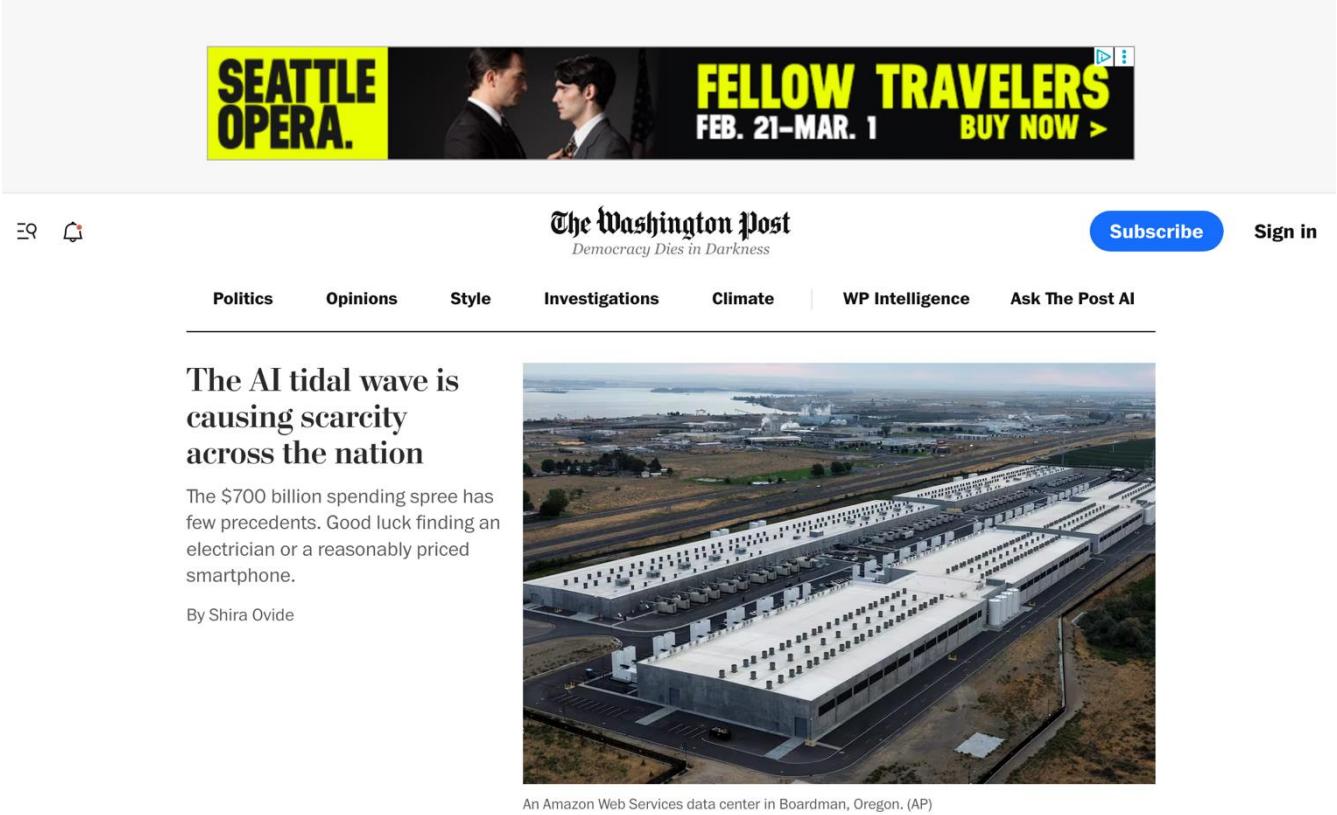
The Importance of Anchor Text

- Anchor text
  - Link label
  - Visible, clickable text in a hyperlink.
  - The words contained in the Anchor text can determine the ranking that the page will receive by search engines.

```
<a href="http://en.wikipedia.org/wiki/Main_Page">Wikipedia</a>
```

FOSTER SCHOOL OF BUSINESS

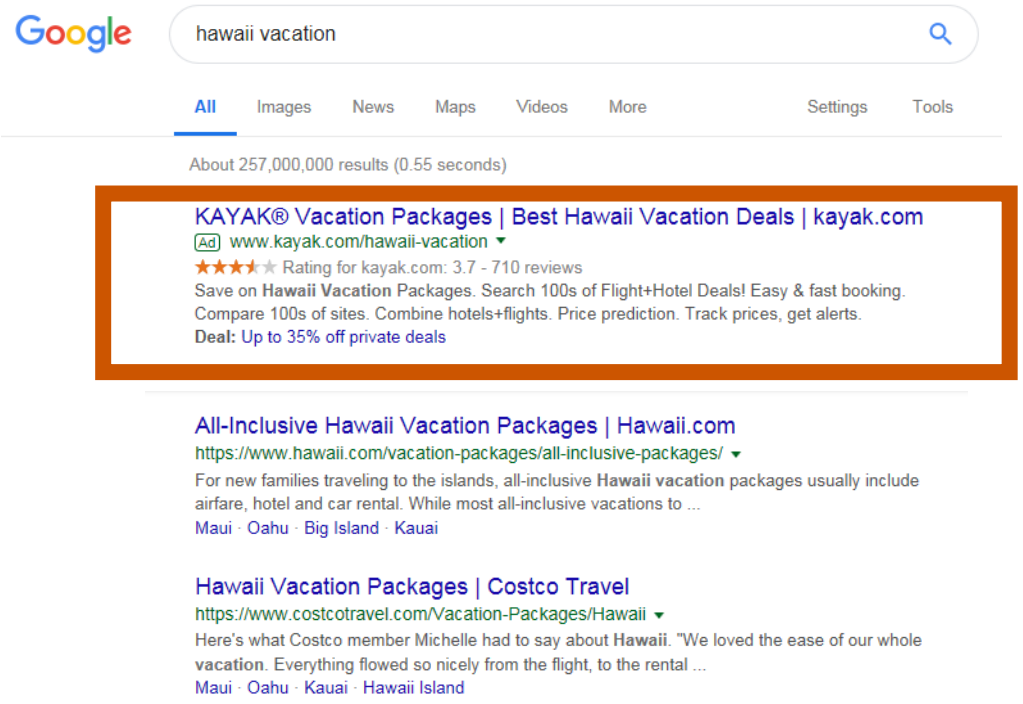
Pricing Method for Ads: Page Impression



- Based on Page Impression
  - Page Impression: the number of times a page has been viewed
  - Traditional pricing

https://advertising.amazon.com/library/guides/display-ads-examples#1

Pricing Method for Ads : Click Through

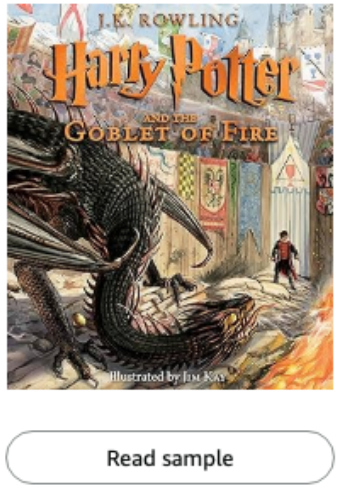


- Based on Click Through
  - Payment based the number of click through
  - Using Cost-Per-Click: from cents to several bucks
  - Less risk

FOSTER SCHOOL OF BUSINESS

Selling Strategy: Fixed Price

- Fixed Price
  - Price as labeled
  - One price for one product/service
  - Estimable cost and valuation



Harry Potter and the Goblet of Fire: The Illustrated Edition (Harry Potter, Book 4) (4) Hardcover – Illustrated, October 8, 2019

by J. K. Rowling (Author), Mr. Jim Kay (Illustrator)

4.9 ★★★★★ 27,402 ratings

4.6 on Goodreads 3,919,866 ratings

Book 4 of 5: Harry Potter Illustrated Editions

See all formats and editions

The fourth book in the beloved Harry Potter series, now illustrated in glorious full color by award-winning artist Jim

Hardcover
$29.98 (Earn 30 pts)
✓prime

Other Used, New, Colle

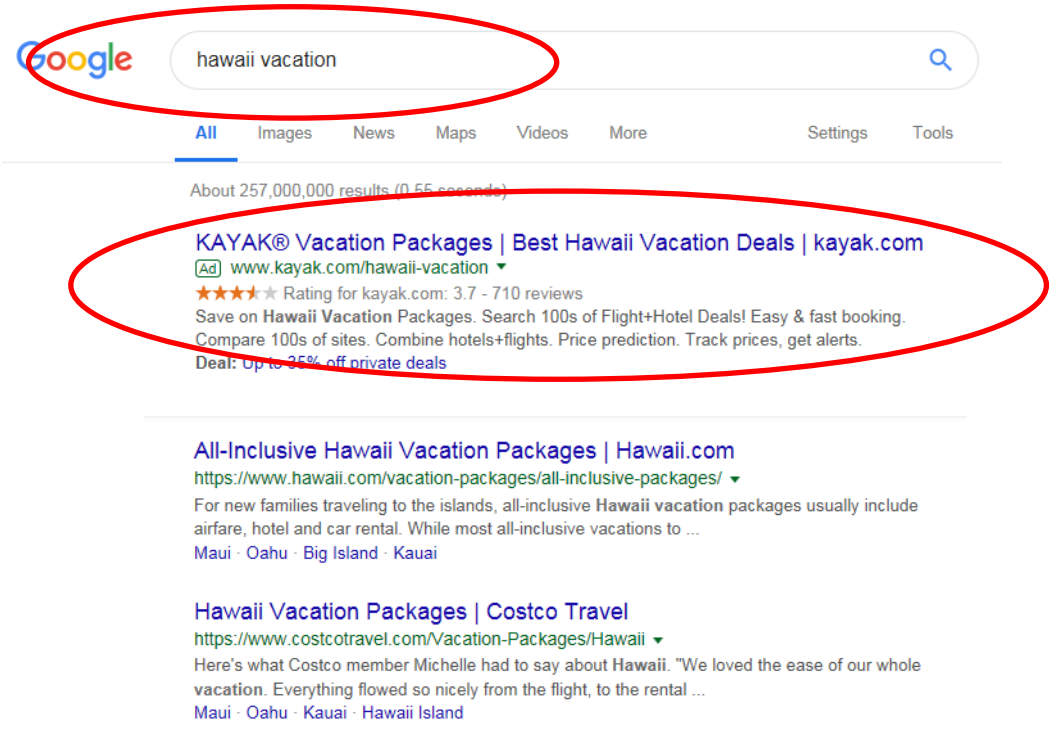
Delivery

Buy new:

-38% $29.98

List Price: ~~$47.99~~

Selling Strategy: Auction



- Dilemma in selling
  - Too many keywords
  - Unpredictable valuation
- Google's bread and butter: using keyword auctions to sell ads.

FOSTER SCHOOL OF BUSINESS

Online Advertisement Metrics

- **CPC/PPC**: Cost per Click/Pay per click
  - Price an advertiser pays for each click on their ad
- **CPM**: Cost per thousand impressions
- **CPA**: Cost Per Action
- **Click Through**: count of the number of people who visit one site and use an ad to get to another
- **CTR**: Click Through Rate
  - Percentage of clicks to impressions
- **Conversion Rate**: percentage of potential customers who actually buy something
- **Daily Budget**
  - Amount advertiser is willing to spend, each day, on PPC ads

FOSTER SCHOOL OF BUSINESS

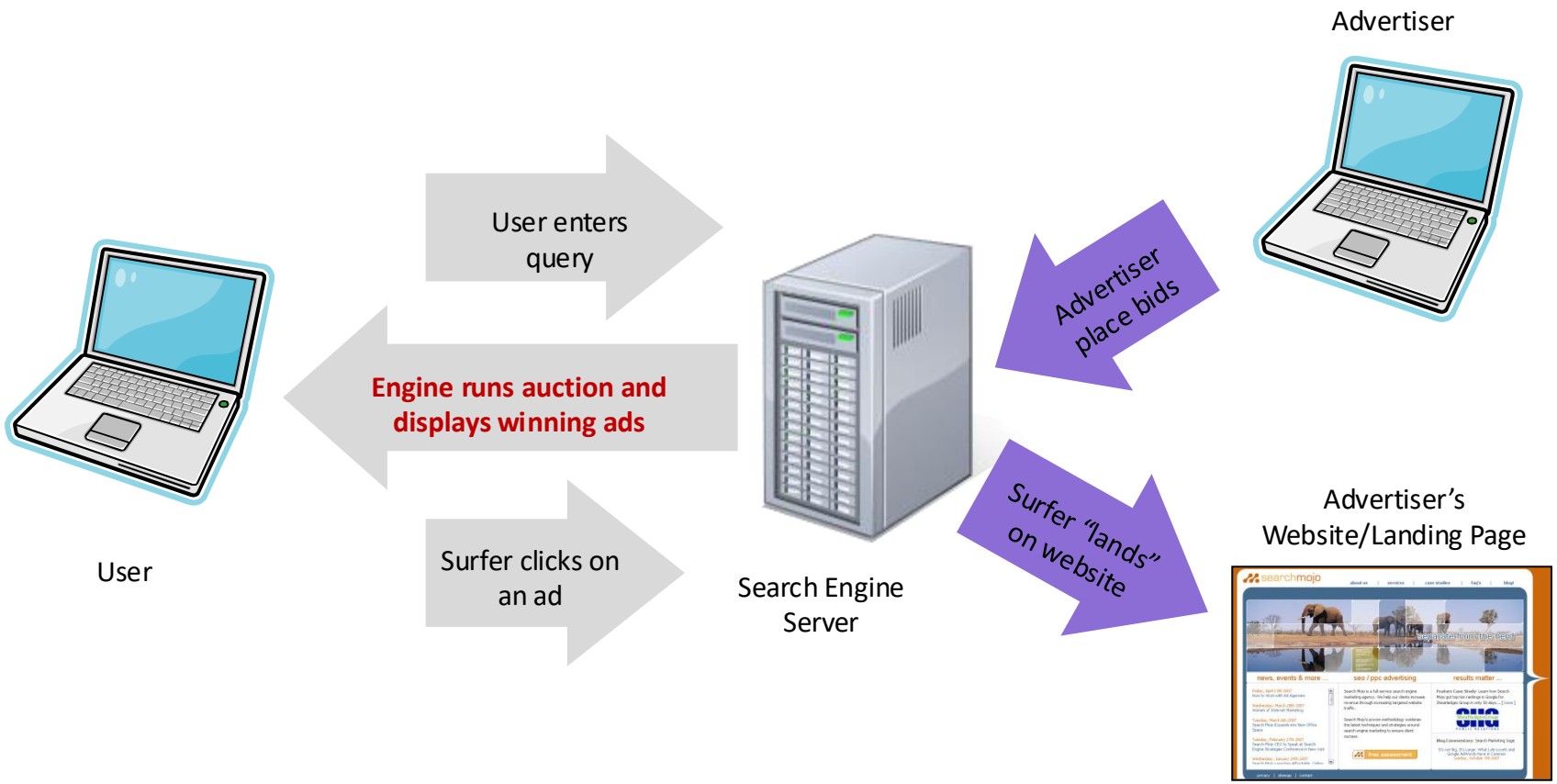
Online Advertising Cost Example

- Q: You are selling advertising space on your website to Amazon.com. The average number of visitors to your website is 1,000,000 per month, and from past experience, you know that your click through rate (CTR) is 1% of which 10% result in actual sales (10% of 1% = 0.1%). Amazon.com has two affiliate programs. In the first affiliate program you get a Cost Per Click (CPC) of ¢5 and in the second affiliate program you get a Cost Per Action (CPA) of ¢25. Which affiliate program will you choose and why? Which affiliate program is more profitable from Amazon.com's point of view?

- Q: Would Amazon.com offer you a Cost per 1000 Impressions (CPM) of $1? Why or why not? Would you prefer this option?

FOSTER SCHOOL OF BUSINESS

How PPC Works



Millions of auctions executed every day

Once the daily budget is reached, the ad will no longer be shown.

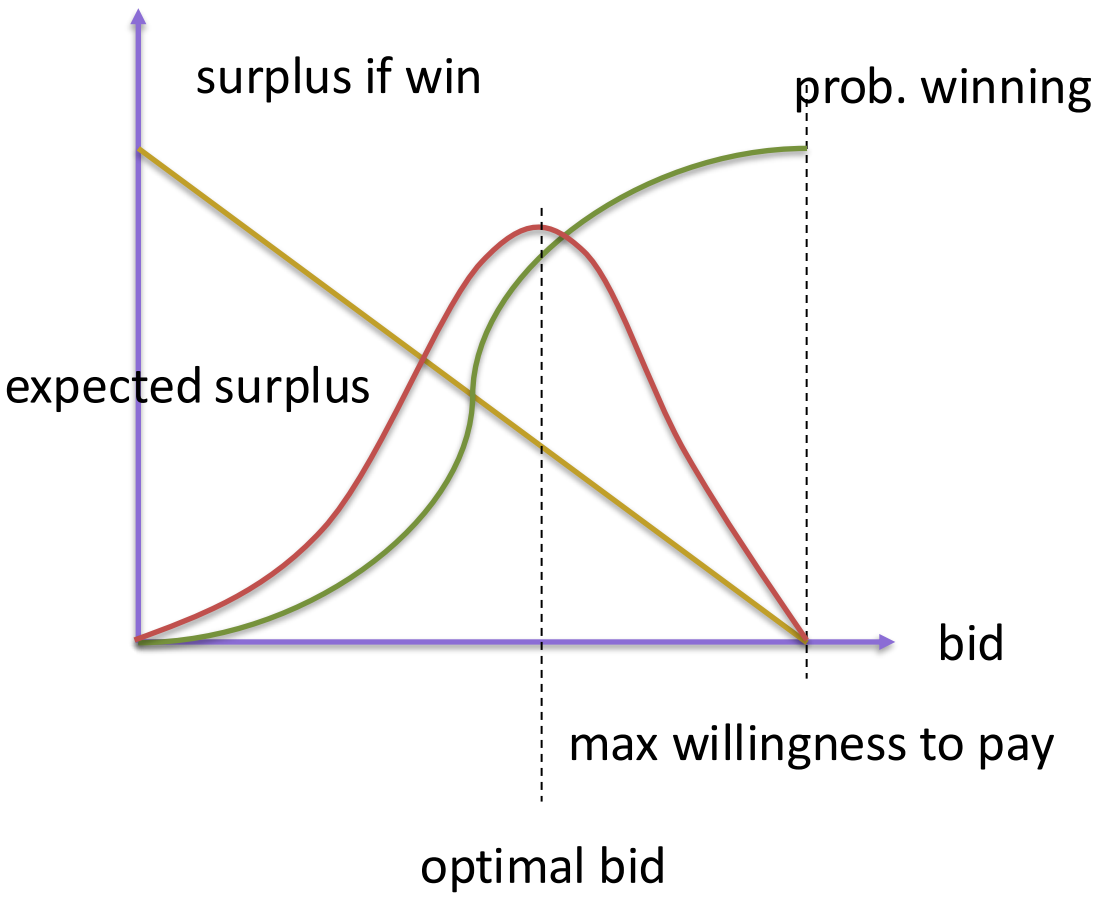
Auctions

- An auction is a process of buying and selling goods or services by offering them up for bids, taking bids, and then selling the item to the highest bidder.
- Auctions are reported to have begun as early as 500 BC when marriage and slave auctions were carried out
  - By the 1500s, information on auctions were documented in France, the Netherlands, and China for slaves, estate auctions and fine art

FOSTER SCHOOL OF BUSINESS

Keyword Auctions: PPC-Based

- Basic model: First-Price Auctions
  - Highest bidder gets highest slot
  - Winner pays winning bid
  - Not ideal! Why?
  - Early Yahoo model
- Second-Price Auction
- Generalized Second-Price Auctions
  - Google's innovation



Standard Auctions

- Suppose advertisers {1, 2, 3} bid CPC $4, $2, and $1, respectively
- Two slots
- Under simple first-price auctions

Advertiser	Bid	Slot Assigned	CPC
1	$4	1	$4
2	$2	2	$2
3	$1	--	--

FOSTER SCHOOL OF BUSINESS

What about CTRs?

- Under simple second-price auctions

Advertiser	Bid	Slot Assigned	CPC
1	$4	1	$2
2	$2	2	$1
3	$1	--	--

- Does Google know something about CTR?
- Implications?
- E.g., CTRs are 0.2%, 0.5%, 2%, respectively?

FOSTER SCHOOL OF BUSINESS

Ad Quality Score

- Search engine
  - needs repeat customers: improve user experience for long term
  - maximizes their revenue
- Ranking based on CPC bids and Quality Score
  - (Bid $ amount) x *(Ad Quality Score)*

If we simplify for demonstration (in practice expected CTR, ad relevance and landing page experience):

~ (Bid $ amount) x (CTR rate)

In other words, rank by expected revenue for Google

- High quality ads → Lower bid amount

Details for quality score: https://support.google.com/google-ads/answer/6167118

Calculation

AdRank = Max CPC x Quality Score

Actual CPC = (AdRank to hit /Quality Score of Advertiser) + $.01

Advertiser	Max CPC	Quality Score	Ad rank	Position	Actual CPC
A	$0.40	6	$0.40 x 6 = 2.4	1	(1.8/6) + $.01 = $0.31
B	$0.35	5	$0.35 x 5 = 1.8	2	(1.6/5) + $.01 = $0.33
C	$0.80	2	$0.80 x 2 = 1.6	3	(1.0/2) + $.01 = $0.51
D	$1.00	1	$1.00 x 1 = 1.0	4	(0.3/1) + $.01 = $0.31
E	$0.05	5	$0.05 x 5 = 0.3	5	(0.2/5) + $.01 = $0.05
F	$0.10	2	$0.10 x 3 = 0.2	6	Minimum Bid

FOSTER SCHOOL OF BUSINESS

Search Types and User Intent

- User intent (When someone types a query, what are they trying to do?):
  - Informational (I want to learn something)
  - Navigational (I want to get to a specific site/brand)
  - Transactional (I want to buy/sign up)
- It is critical to match your ad and landing page to the user's intent to improve CTR, conversion and quality score

FOSTER SCHOOL OF BUSINESS

User Intent Exercise

- “how to clean white sneakers”
- “book flight seattle to madrid”
- “symptoms of vitamin D deficiency”
- “nike official site”
- “order pizza near me”
- “gmail login”
- “buy noise cancelling headphones”
- “what is a large language model”
- “sign up for adobe creative cloud”
- “mit sloan mba program”

FOSTER SCHOOL OF BUSINESS

Mapping Keywords and Content

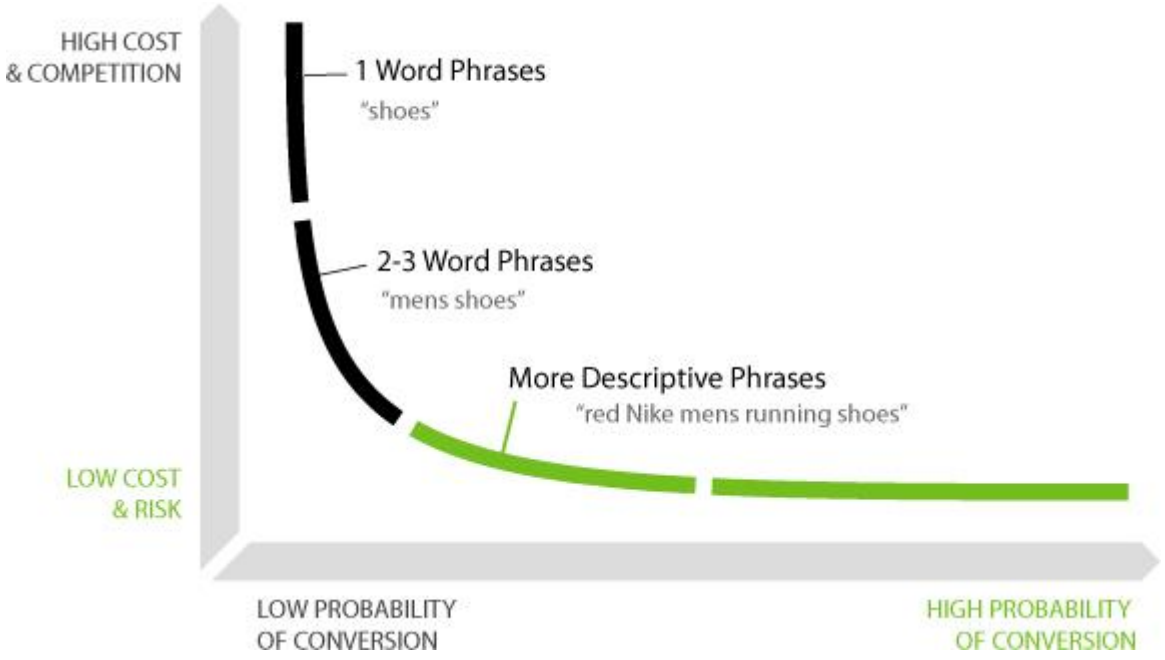
Once we know intent, how do we decide which content (or landing page) should target which queries?

- Start from user intent and topics, not isolated keywords
- For each topic/intent, define:
  - A primary keyword (main target)
  - A cluster of related queries (variants, long-tail, question forms)
- Map each cluster to a specific page or landing experience:
  - SEO: which page should rank for this topic
  - SEM: Which ad group & landing page should serve these queries?

FOSTER SCHOOL OF BUSINESS

Keyword Generation

- Select keywords
  - Head vs. tail
  - Optimize for each page
  - Meta tags



Long-tail keywords are essential for driving targeted traffic and improving SEO performance, especially for new or niche websites.

FOSTER SCHOOL OF BUSINESS

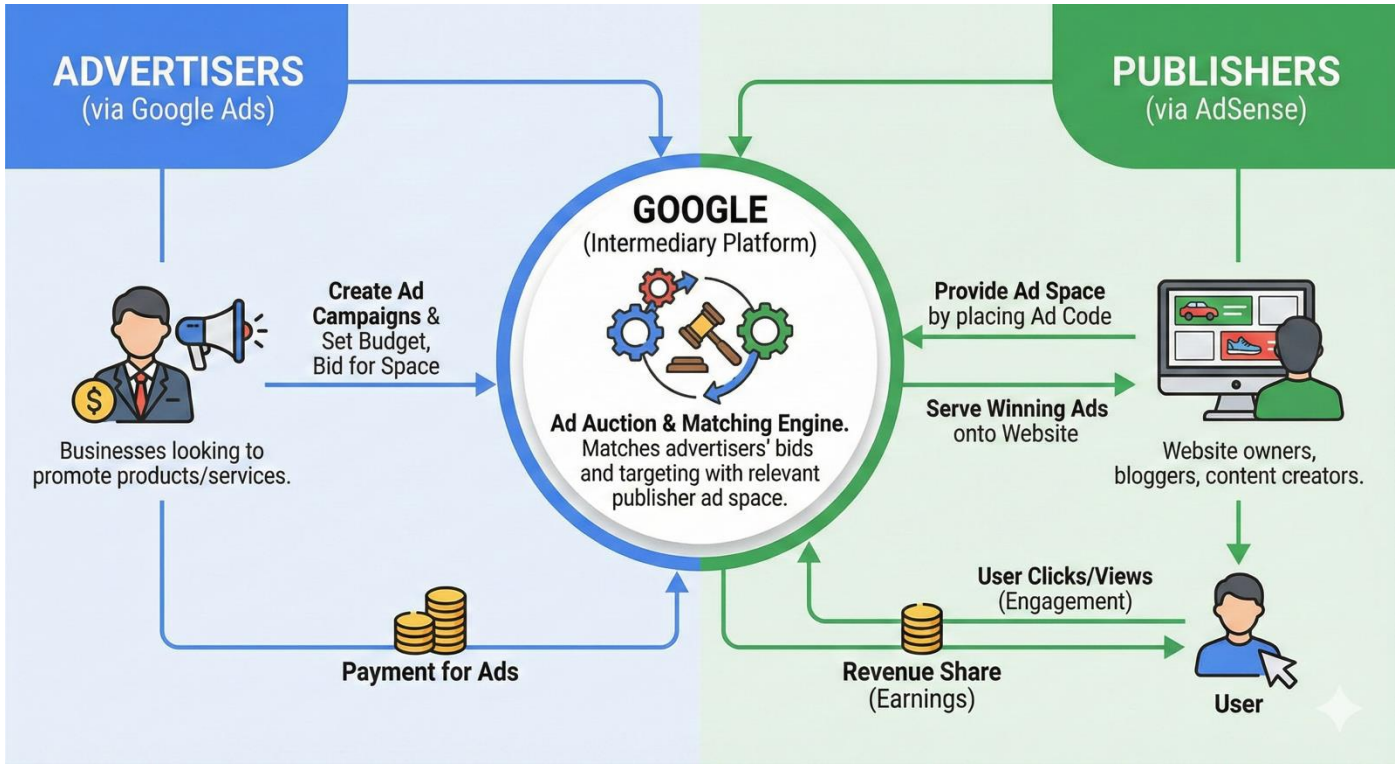
Bidding Considerations and Strategies

- Match types:
  - Exact (very tight control, where you want precision)
  - Phrase (middle ground, ads show for searches that include the meaning of keyword)
  - Broad (most flexible, ads show for searches related to your keywords' meaning)
  - Negative keywords (excluded terms, no ads for irrelevant or low-intent queries, like "free")
- Manual vs. automated:
  - Manual cost-per-click
  - Automated bidding such as targeting CPA/ROAS (we tell Google the business goal and it sets bids in each auction)

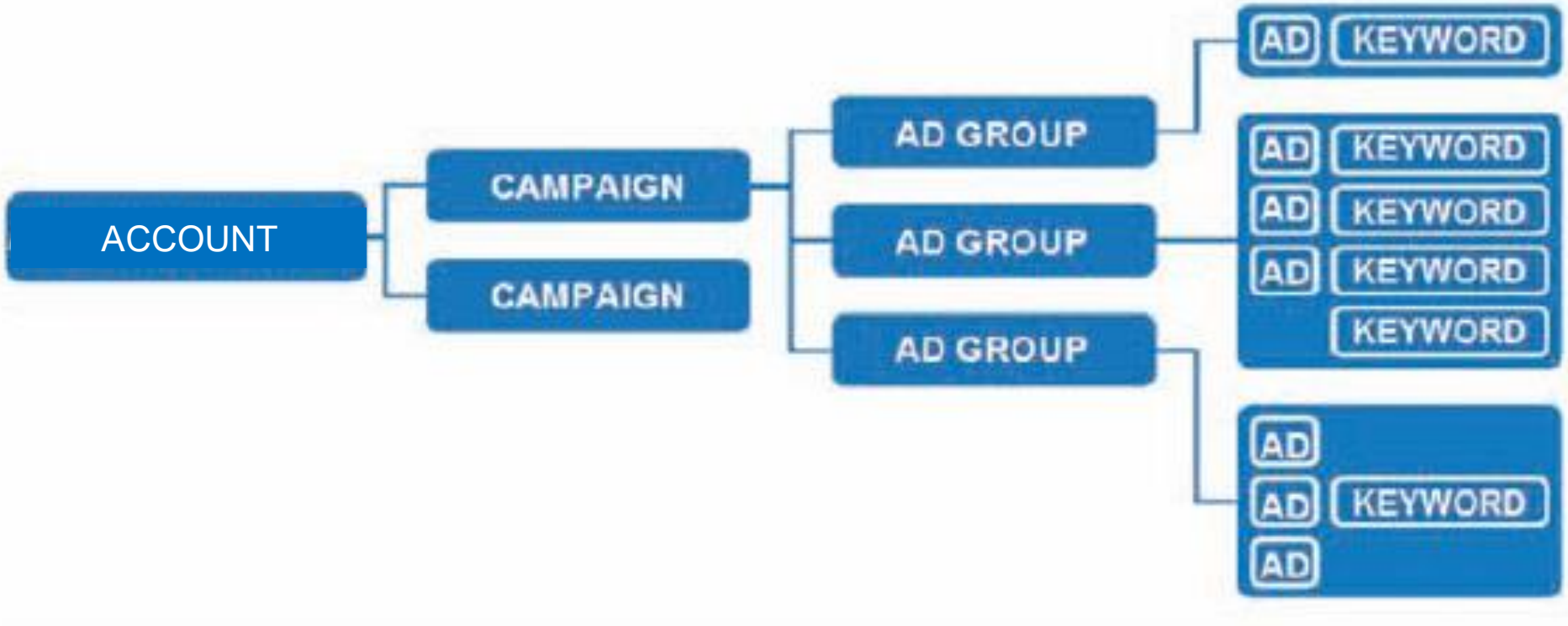
FOSTER SCHOOL OF BUSINESS

Google Advertising Platform

- Google Ads: serves advertisers (to create and run ads)
- AdSense: serves publishers (to display ads and earn revenue)

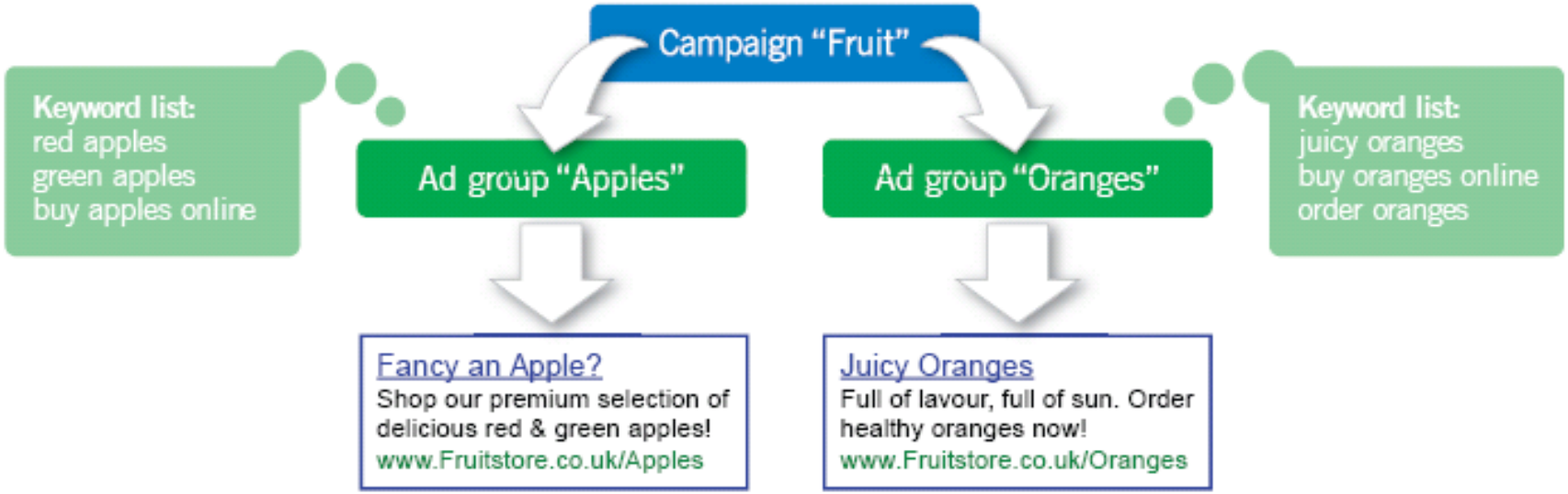


Google Ads Account Structure



FOSTER SCHOOL OF BUSINESS

Campaign Structure



Your Ad Groups, keywords and ad text should be organized in a logical format for the best results

Ad Formats

Comprehensive Insurance | Protect Yourself on a Budget | Get Your Free Quote Today

Ad www.example.com/insurance

Get affordable & trustworthy insurance. 10% discount on all online quotes. Easily compare insurance plans side-by-side in just a few seconds.

Headline (3): 30 characters each

Description (2): 90 characters

Display URL:

- Domain
- Path text (2): 15 characters each

FOSTER SCHOOL OF BUSINESS

Ad Text

- Text should be clear and well-written
- Include keywords in the title
- Include prices and promotions
- Include a call to action
  - Action verbs: buy, order, and purchase
  - Click here
- Avoid using a company name or website domain in the ad text
- Choose a destination URL that points the visitors to a landing page relevant to the keyword entered

FOSTER SCHOOL OF BUSINESS

Examples for Good vs. Bad Ad Text

Headline 1: Online Data Analytics Course
Headline 2: Learn Python, SQL & Real Business Cases

Description: Become job-ready in 12 weeks with live projects, mentor feedback, and career coaching. Start your free intro lesson today.

Display URL: www.exampleuniversity.edu/data-analytics

Headline 1: Top Online Courses
Headline 2: Upgrade Your Skills Today
Description: Choose from hundreds of online programs in tech, business, and more. Enroll now to start learning at your own pace.

Display URL: www.exampleuniversity.edu/courses

FOSTER SCHOOL OF BUSINESS

Google Ads

- Issues to consider
  - Daily budget, Max CPC (affects top placement)
  - Ad generation
  - Tools: Keyword generator, Traffic estimator
  - Using experimentation and A/B testing for better ads
  - Responsive search ads
  - Campaign Settings
    - Ad position preference
    - Targeting capabilities
    - Daytime
    - Geo targeting

From SEM/SEO to AI-Augmented Search Marketing

- Using LLMs to generate and test ad copy and landing page variants
- Search term reports (e.g., from Google Ads/Google Search Console) can have thousands of unique queries:
  - Use BERT embeddings to cluster queries and search terms into meaningful intent groups (label clusters and connect them to metrics by calculating cluster-level KPIs)
  - Use sentiment and topic models on search query reports and landing page behavior to find new opportunities and uncover customer pain points
  - Feed these insights back into campaign structure, ad copy and landing page design
- Example:
  - We discover a BERT cluster of queries about ‘refunds and delays’ with very high bounce rate → we create a dedicated shipping & returns page and adjust ad copy to set expectations honestly.

FOSTER SCHOOL OF BUSINESS